

Fourier method for PDE.
SOLUTIONS TO EXAMPLES

1. We need to solve the wave equation for the string,

$$u_{tt} - c^2 u_{xx} = 0, \quad (1)$$

for the displacement $u(x, t)$, $0 \leq x \leq l$, subject to the boundary conditions

$$u(0, t) = u(l, t) = 0, \quad (2)$$

and initial conditions¹

$$u(x, 0) = 0, \quad u_t(x, 0) = \begin{cases} 0, & 0 < x < a - \varepsilon, \\ v_0, & a - \varepsilon < x < a + \varepsilon, \\ 0, & a + \varepsilon < x < l. \end{cases} \quad (3)$$

We solve Eq. (1) using the variable separation method and seek its solution in the form

$$u(x, t) = v(x)q(t).$$

Applying the boundary conditions (2), we find (see Sec. 2.2 of the notes and Problem sheet 7)

$$u(x, t) = \underbrace{(A \cos \omega_n t + B \sin \omega_n t)}_{q(t)} \underbrace{\sin \frac{n\pi x}{l}}_{v(x)}, \quad (4)$$

where

$$\omega_n = \frac{n\pi c}{l}, \quad n = 1, 2, 3, \dots \quad (5)$$

Using the superposition principle, we construct a more general solution

$$u(x, t) = \sum_{n=1}^{\infty} (A_n \cos \omega_n t + B_n \sin \omega_n t) \sin \frac{n\pi x}{l}, \quad (6)$$

where A_n and B_n are arbitrary coefficients. We find them using the initial conditions.

Setting $t = 0$ in Eq. (6) and using the first of Eq. (3), we have

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l} = 0 \Rightarrow A_n = 0,$$

so that

$$u(x, t) = \sum_{n=1}^{\infty} B_n \sin \omega_n t \sin \frac{n\pi x}{l} \quad (7)$$

Its time-derivative is

$$u_t(x, t) = \sum_{n=1}^{\infty} B_n \omega_n \cos \omega_n t \sin \frac{n\pi x}{l},$$

and setting $t = 0$, we have

$$u_t(x, 0) = \sum_{n=1}^{\infty} B_n \omega_n \sin \frac{n\pi x}{l}. \quad (8)$$

This is a Fourier series with coefficients $B_n \omega_n$, which are found as (Sec. 3.5 of the notes)

$$B_n \omega_n = \frac{2}{l} \int_0^l u_t(x, 0) \sin \frac{n\pi x}{l} dx.$$

¹The condition on the velocity $u_t(x, 0)$ means that a section of length 2ε of the string around $x = a$ is given a velocity v_0 .

Using $u_t(x, 0)$ from Eq. (3), we have

$$\begin{aligned}
B_n \omega_n &= \frac{2}{l} \int_{a-\varepsilon}^{a+\varepsilon} v_0 \sin \frac{n\pi x}{l} dx = \frac{2v_0}{l} \frac{l}{n\pi} \left[-\cos \frac{n\pi x}{l} \right]_{a-\varepsilon}^{a+\varepsilon} \\
&= \frac{2v_0}{n\pi} \left(\cos \frac{n\pi(a-\varepsilon)}{l} - \cos \frac{n\pi(a+\varepsilon)}{l} \right) \\
&= \frac{2v_0}{n\pi} \left(\cos \frac{n\pi a}{l} \cos \frac{n\pi\varepsilon}{l} + \sin \frac{n\pi a}{l} \sin \frac{n\pi\varepsilon}{l} - \cos \frac{n\pi a}{l} \cos \frac{n\pi\varepsilon}{l} + \sin \frac{n\pi a}{l} \sin \frac{n\pi\varepsilon}{l} \right) \\
&= \frac{4v_0}{n\pi} \sin \frac{n\pi a}{l} \sin \frac{n\pi\varepsilon}{l},
\end{aligned}$$

i.e.,

$$B_n \omega_n = \frac{4v_0}{n\pi} \sin \frac{n\pi a}{l} \sin \frac{n\pi\varepsilon}{l}.$$

Using Eq. (5), we have

$$B_n = \frac{4v_0 l}{\pi^2 n^2 c} \sin \frac{n\pi a}{l} \sin \frac{n\pi\varepsilon}{l}. \quad (9)$$

Hence, the solution, Eq. (7), becomes

$$u(x, t) = \frac{4v_0 l}{\pi^2 c} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi a}{l} \sin \frac{n\pi\varepsilon}{l} \sin \frac{n\pi c t}{l} \sin \frac{n\pi x}{l}. \quad (10)$$

The momentum imparted to the string² is $P = 2\varepsilon\rho v_0$, where ρ is the linear mass density of the string and $2\varepsilon\rho$ is the mass of the segment of length 2ε . Using $v_0 = P/2\varepsilon\rho$, we re-write Eq. (10) as

$$u(x, t) = \frac{2Pl}{\pi^2 c \varepsilon \rho} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi a}{l} \sin \frac{n\pi\varepsilon}{l} \sin \frac{n\pi c t}{l} \sin \frac{n\pi x}{l}. \quad (11)$$

We can now take the limit $\varepsilon \rightarrow 0$ (for the string struck at the point $x = a$), and use³

$$\sin \frac{n\pi\varepsilon}{l} \simeq \frac{n\pi\varepsilon}{l} \quad (\text{for } \varepsilon \rightarrow 0),$$

which gives

$$u(x, t) = \frac{2P}{\pi c \rho} \sum_{n=1}^{\infty} \frac{\sin \frac{n\pi a}{l}}{n} \sin \frac{n\pi c t}{l} \sin \frac{n\pi x}{l}. \quad (12)$$

In particular, if the string is struck in the middle, $a = l/2$, then

$$\sin \frac{n\pi a}{l} = \sin \frac{n\pi}{2} = \begin{cases} 0, & n = 2m, \\ (-1)^m, & n = 2m + 1, \end{cases}$$

which means that only odd $n = 2m + 1$ ($m = 0, 1, 2, \dots$) contribute to the sum, so we have

$$u(x, t) = \frac{2P}{\pi c \rho} \sum_{m=0}^{\infty} \frac{(-1)^m}{2m+1} \sin \frac{(2m+1)\pi c t}{l} \sin \frac{(2m+1)\pi x}{l}. \quad (13)$$

2. We need to solve the heat equation for the rod,

$$u_t - K u_{xx} = 0, \quad (14)$$

²The momentum is mass $\times$ velocity.

³Use the Maclaurin expansion, $\sin x = x - \frac{x^3}{3!} + \dots$, and keep only the first-order term, $\sin x \simeq x$ for $|x| \ll 1$.

for the temperature $u(x, t)$, $0 \leq x \leq l$, with insulating boundary conditions⁴

$$u_x(0, t) = u_x(l, t) = 0, \quad (15)$$

and initial conditions

$$u(x, 0) = f(x). \quad (16)$$

Using variable separation for Eq. (14), we seek its solution in the form⁵

$$u(x, t) = v(x)q(t), \quad (17)$$

and have

$$v(x) \frac{dq}{dt} - q(t) K \frac{d^2 v}{dx^2} = 0.$$

Dividing through by $v(x)q(t)$, we obtain

$$\underbrace{\frac{1}{q(t)} \frac{dq}{dt}}_{-\lambda} - \underbrace{\frac{K}{v(x)} \frac{d^2 v}{dx^2}}_{-\lambda} = 0.$$

Denoting the separation constant $-\lambda$, we have

$$\frac{1}{q(t)} \frac{dq}{dt} = -\lambda \quad \Leftrightarrow \quad \frac{dq}{dt} = -\lambda q,$$

which has the solution

$$q(t) = Ce^{-\lambda t}, \quad (18)$$

where C is an arbitrary constant. The coordinate part gives

$$\frac{K}{v(x)} \frac{d^2 v}{dx^2} = -\lambda \quad \Leftrightarrow \quad \frac{d^2 v}{dx^2} + \frac{\lambda}{K} v = 0.$$

Its general solution is

$$v(x) = A \cos \sqrt{\frac{\lambda}{K}} x + B \sin \sqrt{\frac{\lambda}{K}} x.$$

Since $u_x(x, t) = \frac{dv}{dx} q(t)$, the boundary conditions (15) mean that $v'(0) = v'(l) = 0$. Using

$$\frac{dv}{dx} = -A \sqrt{\frac{\lambda}{K}} \sin \sqrt{\frac{\lambda}{K}} x + B \sqrt{\frac{\lambda}{K}} \cos \sqrt{\frac{\lambda}{K}} x, \quad (19)$$

for $x = 0$, we have $B \sqrt{\lambda/K} = 0$, which gives $B = 0$, so that

$$v(x) = A \cos \sqrt{\frac{\lambda}{K}} x. \quad (20)$$

Using $B = 0$ and $v'(l) = 0$ in Eq. (19), gives

$$-A \sqrt{\frac{\lambda}{K}} \sin \sqrt{\frac{\lambda}{K}} l = 0 \quad \Rightarrow \quad \sqrt{\frac{\lambda}{K}} l = n\pi, \quad n = 0, 1, 2, \dots$$

Therefore, $\sqrt{\frac{\lambda}{K}} = \frac{n\pi}{l}$, and we have⁶

$$\lambda = \frac{n^2 \pi^2}{l^2} K, \quad n = 0, 1, 2, \dots, \quad (21)$$

and Eq. (17) gives

$$u(x, t) = A \cos \frac{n\pi x}{l} e^{-n^2 \pi^2 K t / l^2}. \quad (22)$$

⁴The Fourier law for the heat flux density, $j = -\kappa \partial u / \partial x$ means that $j = 0$ is equivalent to $\partial u / \partial x = 0$.

⁵These steps are similar to those taken in Sec. 2.4 of the notes.

⁶Note that $n = 0$ gives $\lambda = 0$, but the corresponding solution (20) is nonzero.

Using the superposition principle, we construct a more general solution

$$u(x, t) = \sum_{n=0}^{\infty} A_n \cos \frac{n\pi x}{l} e^{-n^2\pi^2 Kt/l^2}. \quad (23)$$

This solution must satisfy the initial condition (16), which gives

$$\sum_{n=0}^{\infty} A_n \cos \frac{n\pi x}{l} = f(x).$$

We see that A_n are the coefficients of the cosine Fourier series. Its standard form is

$$\frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi x}{l} = f(x),$$

and the coefficients are given by

$$A_0 = \frac{2}{l} \int_0^l f(x) dx, \quad A_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx.$$

We can substitute these into Eq. (23), changing the integration variable from x to ξ , to avoid confusion, and keeping in mind that the coefficient for the $n = 0$ term is $A_0/2$:

$$u(x, t) = \frac{1}{l} \int_0^l f(\xi) d\xi + \frac{2}{l} \sum_{n=1}^{\infty} \left(\int_0^l f(\xi) \cos \frac{n\pi \xi}{l} d\xi \right) A_n \cos \frac{n\pi x}{l} e^{-n^2\pi^2 Kt/l^2}. \quad (24)$$

It is interesting to note that all terms in the sum $\sum_{n=1}^{\infty}$ tend to zero as $t \rightarrow \infty$, since $\lim_{t \rightarrow \infty} e^{-n^2\pi^2 Kt/l^2} = 0$. Hence, in the large- t limit, the temperature tends to a constant,

$$u(x, t) \rightarrow \frac{1}{l} \int_0^l f(\xi) d\xi. \quad (25)$$

It is easy to recognise that the expression on the right-hand side is the average initial temperature in the rod. Since the ends of the rod are insulated, the heat energy (which is proportional to the temperature) distributes uniformly along the rod as the time goes by, leading to the long-time equilibrium (25)⁷.

3. Let us consider a string of length l , $0 \leq x \leq l$, with fixed ends,

$$u(0, t) = u(l, t) = 0, \quad (26)$$

initially at rest in equilibrium,

$$u(x, 0) = 0, \quad u_t(x, 0) = 0, \quad (27)$$

to which a uniform load q is applied at $t = 0$ ⁸. For $t > 0$, the displacement $u(x, t)$ satisfies the equation

$$u_{tt} - c^2 u_{xx} = \frac{q}{\rho}. \quad (28)$$

This equation is inhomogeneous. Let us seek its solution in the form

$$u(x, t) = \tilde{u}(x, t) + u_s(x), \quad (29)$$

and require that the time-independent term $u_s(x)$ satisfies Eq. (28):

$$-c^2(u_s)_{xx} = \frac{q}{\rho}, \quad (30)$$

⁷By contrast, if the boundary conditions were $u(0, t) = u(l, t) = 0$, i.e., the ends kept at zero temperature, the large-time limit of the temperature would be $u(x, t) \rightarrow 0$.

⁸Here q is the force per unit length of the string.

so that

$$\frac{d^2 u_s}{dx^2} = -\frac{q}{c^2 \rho}.$$

Integrating this equation, we have

$$\frac{du_s}{dx} = -\frac{qx}{c^2 \rho} + C_1,$$

and

$$u_s(x) = -\frac{qx^2}{2c^2 \rho} + C_1 x + C_2.$$

The constants C_1 and C_2 can be found from the boundary conditions $u_s(0) = u_s(l) = 0$, which give

$$0 + 0 + C_2 = 0 \quad \text{and} \quad -\frac{ql^2}{2c^2 \rho} + C_1 l + C_2 = 0.$$

We then have $C_2 = 0$ and $C_1 = \frac{ql}{2c^2 \rho}$, which gives

$$u_s(x) = -\frac{ql^2}{2c^2 \rho} + \frac{ql}{2c^2 \rho} x,$$

or

$$u_s(x) = \frac{q}{2c^2 \rho} x(l - x). \quad (31)$$

Substituting $u(x, t)$ from Eq. (29) into Eq. (28), and using Eq. (30), we obtain

$$\tilde{u}_{tt} - c^2 \tilde{u}_{xx} = 0,$$

i.e., $\tilde{u}(x, t)$ satisfies the homogeneous wave equation with boundary conditions $\tilde{u}(0, t) = \tilde{u}(l, t) = 0$. The solution to this problem was obtained in Example 1, Eq. (6), i.e., we have

$$\tilde{u}(x, t) = \sum_{n=1}^{\infty} (A_n \cos \omega_n t + B_n \sin \omega_n t) \sin \frac{n\pi x}{l}, \quad (32)$$

with $\omega_n = n\pi c/l$, $n = 1, 2, 3, \dots$. Combining this with $u_s(x)$ from Eq. (31) gives

$$u(x, t) = \frac{q}{2c^2 \rho} x(l - x) + \sum_{n=1}^{\infty} (A_n \cos \omega_n t + B_n \sin \omega_n t) \sin \frac{n\pi x}{l}. \quad (33)$$

This solution must satisfy the initial conditions (27). Setting $t = 0$, we obtain

$$\frac{q}{2c^2 \rho} x(l - x) + \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l} = 0. \quad (34)$$

Taking the time derivative of $u(x, t)$, Eq. (33), and setting $t = 0$ gives

$$\sum_{n=1}^{\infty} B_n \omega_n \sin \frac{n\pi x}{l} = 0. \quad (35)$$

This equation yields $B_n = 0$, while Eq. (34) gives

$$\sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l} = -\frac{q}{2c^2 \rho} x(l - x).$$

The coefficients A_n of this Fourier series on $(0, l)$ are found from the standard formula

$$A_n = \frac{2}{l} \int_0^l \left[-\frac{q}{2c^2 \rho} x(l - x) \right] \sin \frac{n\pi x}{l} dx,$$

or

$$A_n = -\frac{q}{c^2 \rho l} \int_0^l x(l - x) \sin \frac{n\pi x}{l} dx. \quad (36)$$

The integral on the right-hand side is found using integration by parts twice:

$$\begin{aligned}
\int_0^l x(l-x) \sin \frac{n\pi x}{l} dx &= \underbrace{\left[x(l-x) \left(-\frac{l}{n\pi} \right) \cos \frac{n\pi x}{l} \right]_0^l}_{=0} + \frac{l}{n\pi} \int_0^l \cos \frac{n\pi x}{l} (l-2x) dx \\
&= \frac{l^2}{n\pi} \int_0^l \cos \frac{n\pi x}{l} dx - \frac{2l}{n\pi} \int_0^l x \cos \frac{n\pi x}{l} dx \\
&= \frac{l^2}{n\pi} \frac{l}{n\pi} \underbrace{\left[\sin \frac{n\pi x}{l} \right]_0^l}_{=0} - \frac{2l}{n\pi} \frac{l}{n\pi} \underbrace{\left[x \sin \frac{n\pi x}{l} \right]_0^l}_{=0} + \frac{2l}{n\pi} \frac{l}{n\pi} \int_0^l \sin \frac{n\pi x}{l} dx \\
&= \frac{2l^2}{n^2\pi^2} \left[-\frac{l}{n\pi} \cos \frac{n\pi x}{l} \right]_0^l \\
&= \frac{2l^3}{n^3\pi^3} [-\cos n\pi + \cos 0] \\
&= \frac{2l^3}{n^3\pi^3} [1 - (-1)^n],
\end{aligned}$$

where we also used the fact that $\sin n\pi = \sin 0 = 0$ and $\cos n\pi = (-1)^n$. Hence, we have from Eq. (36)

$$A_n = -\frac{q}{c^2\rho l} \frac{2l^3}{n^3\pi^3} [1 - (-1)^n] = -\frac{2ql^2}{c^2\rho n^3\pi^3} [1 - (-1)^n].$$

This coefficient is zero for even n , and nonzero for odd $n = 2m + 1$, when is given by

$$A_n = -\frac{4ql^2}{c^2\rho(2m+1)^3\pi^3}.$$

Substituting this coefficient and $B_n = 0$ into Eq. (33), and changing the summation index from n to $m = 0, 1, 2, \dots$, we obtain the final answer in the form

$$u(x, t) = \frac{q}{2c^2\rho} \left[x(l-x) - \frac{8l^2}{\pi^3} \sum_{m=0}^{\infty} \frac{1}{(2m+1)^3} \cos \frac{(2m+1)\pi ct}{l} \sin \frac{(2m+1)\pi x}{l} \right].$$